

Analytical Problem Solving.

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1.
(a)

Operations :

- ⇒ Append (10)
- ⇒ Append (20)
- ⇒ Append (15)

Final Linked List :

head → 10 → 20 → 15 → NULL ^ current

b)

Operations :

- ⇒ Append (10)
- ⇒ Append (20)
- ⇒ Append (15)

10 → 20 → 15

Move to start ()

current

↓

10

↓

20

↓

15

Insert (39)

39 → 10 → 20 → 15 ^ current

Next ()

39 → 10 → 20 → 15

^

current

Insert (12)

Final List :

39 → 12 → 10 → 20 → 15 → NULL

^

current

2. Binary Search

Array :

66 111 212 323 482 543 643 649 777 888

Algorithm :-

1. Set $Low = 0$, $High = n - 1$
2. Find $Mid = (Low + High) / 2$
3. If $key = Mid$ element $\rightarrow$ Found
4. If $key < Mid$ element $\rightarrow$ search Left
5. Else search Right
6. Repeat until found or $Low > High$

Case 1

Key = 111

Output :

Element found
Position = 2

Case 2

key = 888

Output :

Element found
position = 10

Case 3

Key = 999

Output :

Element Not found

Time Complexity :-

- $\rightarrow$ Best Case = $O(1)$
- $\rightarrow$ Average Case = $O(\log n)$
- $\rightarrow$ Worst Case = $O(\log n)$

3. Stack (LIFO)

Stack size = 5

Operations:

Push (45)

45

Push (22)

22

45

Push (11)

11

22

45

Push (77)

77

11

22

45

Pop()

77 removed

11

22

45

Push (44)

44

11 (77)

22

45

Push (88)

88

44

11

22

45

Stack Full

Pop()

88 removed

44

11

22

45

Pop()

44 removed

11

22

45

Pop()

11 removed

Final stack:

22

45

Top element = 22

Top Index = 1 (0-based indexing)

4. Expression Conversion

Expression:

$((a+b)+c*(d+e)+f)*(g+h)$

Prefix

$* + + + ab * c + de f + gh$

Postfix

$$ab + cde + * + f + gh + *$$

5.

Queue vs Priority Queue

Feature	Queue	Priority Queue
Order	FIFO	priority based
Fairness	High	Medium
Waiting time	Same for all	VIP gets less waiting
Insertion	$O(1)$	$O(\log n)$
Deletion	$O(1)$	$O(\log n)$
Best Use	Normal ticket counter	Hospital, Airport, Emergency services

Comparison :-

Simple Queue

- Follows First-in-First-out (FIFO)
- All customers are treated equally
- Easy to implement

Priority Queue

- VIP customers are served first
- Reduces waiting time for high-priority customers
- May increase waiting time for normal customers
- More computationally expensive than a Simple Queue